Entire

§ **Entire**

Parameters

+ $\mathrm{A} \in \mathsf{Set}$
+ $\mathrm{B} \in \mathsf{Set}$
+ $\mathrm{R} \in A \sim B$

Assumptions

a. Reflexive (Kernel R)

Observations

1. $\mathrm{id}_{\mathrm{B}} \subseteq R^{\circ} \cdot R$
2. $(\forall x \mid x \in B : (\exists a \mid a \in A : a \mathrel{\mathrm{R}} x))$

■

Proof of Observation 1.

◆ Entire A B R

≡ Definition of Entire

Reflexive (Kernel R)

≡ Definition of Reflexive

$\mathrm{id}_{\mathrm{B}} \subseteq$ Kernel R

≡ Definition of Kernel

$\mathrm{id}_{\mathrm{B}} \subseteq \mathrm{R}^{\circ} \cdot R$

■

Proof of Observation 2.

◆ $\mathrm{id_B} \subseteq R^\circ \cdot R$

$\equiv$ Extensionality

$(\forall y, x \mid y \, \mathrm{id_B} \, x : y(R^\circ \cdot R)x)$

$\equiv$ Identity

$(\forall y, x \mid y = x \wedge x \in B : y(R^\circ \cdot R)x)$

$\equiv$ One-point rule

$(\forall x \mid x \in B : x(R^\circ \cdot R)x)$

$\equiv$ Composition

$(\forall x \mid x \in B : (\exists a \mid : x \, R^\circ \, a \wedge a \, R \, x))$

$\equiv$ Converse

$(\forall x \mid x \in B : (\exists a \mid : a \, R \, x \wedge a \, R \, x))$

$\equiv$ Converse

$(\forall x \mid x \in B : (\exists a \mid : a \, R \, x))$

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